Week 7 Notes & Quiz

Notes for all six Week 7 lectures, made only from the 159 slides in *WEEK 7.pdf*. Each yellow tag such as S104 is the slide (PDF page) the point comes from. The 50-question practice questions and answer key are at the end.

STOP	LECTURE	SLIDES
L31	**Supply Chain Impellers and Value Chain**	S1–S22
L32	**Strategic Supply Chain Management**	S23–S52
L33	**Strategic Supply Chain Management (contd.): SWOT and Porter**	S53–S81
L34	**Achieving Supply Chain Strategic Fit**	S82–S99
L35	**Achieving Supply Chain Strategic Fit (contd.)**	S100–S133
L36	**Make or Buy Decision**	S134–S159
+	**Cheat sheet · Slide quirks · Quiz (50 Qs)**	—

LECTURE 31 S1–S22

Supply Chain Impellers and Value Chain

- **Impellers** are the driving forces that propel supply chain performance (facilities, inventory, transport, information, sourcing, pricing and more). S3

IMPELLER	MEANING
Traditional three S4	
Empowered customer	Knowledgeable, demanding: more variety, availability, competitive prices; transparency (like seeing restaurant kitchens).
Developments in IT tools	Real-time tracking and visibility; tools: **MRP, ERP, VMI, TAV**.
Globalization	New markets and consumers, more competition and risk.
Two more in some literature S5	
Focus on sustainability	Ethical sourcing, energy-efficient logistics, waste reduction.
Collaboration and integration	Across internal functions and with external partners.

Four supply chain concepts

Systems concept	The chain is one interconnected system; optimize the whole, not segments; disruptions ripple. S8
Total cost concept	Count all costs, direct, indirect and hidden, and minimize the total. S9
Trade-off concept	Improving one aspect costs another: cost vs quality, cost vs speed, inventory vs responsiveness. S10
Bullwhip effect	Small consumer demand swings become **amplified orders upstream**. Causes: information delays, forecast errors, order batching. Cures: better forecasting, shorter lead times, **VMI**, information sharing. S11

Value chain in a company

- Primary chain: **New product development → marketing and sales → operations → distribution → service**. S12–S14
- Supporting functions: **finance, accounting, information technology, human resources**. S15

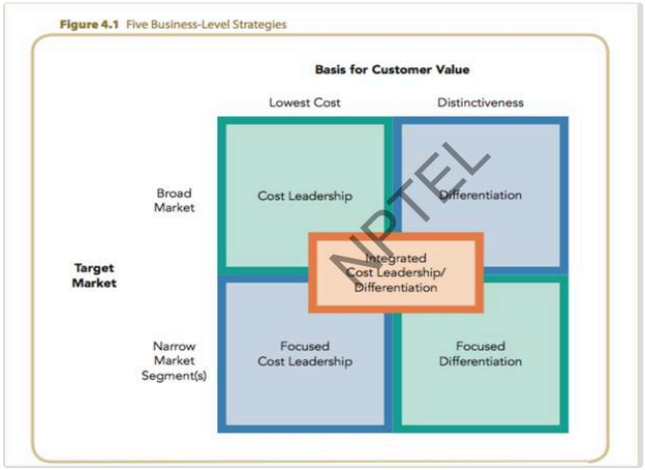
Essential considerations for supply chain success 5 items

1. **Supply sense**: anticipate and understand changes in the supply environment. S17
2. **Supply response**: operations that make things happen (manufacturing, asset management); adjust supply to change. S17
3. **Deciding and committing**: strategic decisions aligned with business goals. S18
4. **Demand sense**: sense and anticipate changes in customer demand with data and analytics. S18
5. **Demand response**: react quickly by adjusting production, inventory, distribution. S19

LECTURE 32 S23–S52

Strategic Supply Chain Management

- A **supply chain strategy** is an overarching plan for planning, design, execution, control and monitoring of supply chain activities; it must align with business goals. Best-in-class firms have a separate strategy for make-to-stock, make-to-order and engineer-to-order. S25
- Strategic management process: review the current situation (mission, objectives) → analyse internal and external environment (SWOT) → develop strategies (corporate, business, functional) → implement → evaluate. Formulation = creating; implementation = putting into action. S26
- Components of SC strategy: **sourcing, logistics, production, distribution, customer service**. S28
- 21st-century competition: rapid technological change, global economy, globalization, importance of knowledge and people. Competitive advantage needs deciding both what the firm **will do** and what it **will not do**. S30–S31



Original slide S32, five business-level strategies. Broad market: **cost leadership** (lowest cost) or **differentiation** (distinctiveness). Narrow segment: **focused cost leadership** or **focused differentiation**. Middle: **integrated cost leadership/differentiation**.

- Cost advantage: control cost drivers and reconfigure the value chain (new process, raw material, automation, forward or backward integration, new channel, relocation, direct sales). S33–S34
- Differentiation advantage: lower buyers' costs, raise performance, create perceptions of uniqueness and reluctance to switch. S35–S36
- Effective SC strategy is **collaborative, agile, resilient, digital**, and **invests in cybersecurity**. S37–S38
- Optimization: S&OP, workforce management, network design, demand planning, inventory optimization. S39
- Strategy per value chain component: product development (portfolio, in-house vs outsourced), marketing (segmentation, positioning, pricing, promotion), supply chain (procurement, operations, distribution, outsourcing), design and service strategies. S41–S46

Cases

- **Unilever (CPFR)**: five-year plan from early 2000; savings of **$14.24 billion**; merged almost **30 warehouses into 5** hubs; **Collaborative Planning, Forecasting and Replenishment** with retailers raised forecast accuracy **10%**, cut inventory holding **10%**, raised sales volume **5%**. S47–S48
- **Eaton**: resilience in the pandemic via education, multi-sourcing, supplier collaboration. **Intel**: ethical sourcing with a **smelter audit protocol** and RFID so tin, tungsten, gold and tantalum are conflict-free. S49

LECTURE 33 S53–S81

Strategic SCM (contd.): SWOT and Porter

- External environment → **opportunities and threats**: what the firm **might** do. Internal environment → unique resources, capabilities, competencies: what the firm **can** do. S56–S57
- Internal analysis: resources (tangible, intangible) → capabilities → core competencies → competitive advantage → strategic competitiveness. Four criteria of sustainable advantage: **valuable, rare, costly to imitate, non-substitutable**. S58
- Value is measured by product performance and attributes customers pay for; superior value → above-average returns. Walmart: "everyday low pricing". S59

SWOT

- Strengths and weaknesses are **internal**; opportunities and threats are **external**. S61–S63
- Supply chain SWOT example: strengths = purchasing power, payment history, quality, core competencies; weaknesses = poor credit, late deliveries, capacity limits; opportunities = more warehouse space, new software; threats = downturns, supplier bankruptcies, rising prices. S63
- **Amazon**: fleet, warehouse network, technology. **2,373+** facilities worldwide, **200+** fulfilment centres in North America, about **319 million sq ft** of US warehousing. **FBA** (Fulfilled by Amazon), FBA Onsite, robots, **Scout** six-wheeled delivery robot. Inventory turnover about **8.39** (FY Dec 2022). S64–S65
- Amazon SWOT: strengths brand, pioneer, cost advantage; weaknesses low margins, seasonality; threats patent infringement, cybersecurity. S67

Porter's Five Forces

- Created by Harvard's **Michael Porter**, published **1979** ("How Competitive Forces Shape Strategy"). S68
- Applied to supply chains: **supplier power** (concentration, switching costs), **buyer power** (concentration, price sensitivity), **threat of new entrants** (barriers, technology), **threat of substitutes**, **rivalry among competitors**. S69–S72

Implementing SC strategy

Identify demand types · understand the product portfolio (different strategies per line) · know customers' locations · manage inventory · know suppliers' locations (foreign suppliers → longer lead times). S74–S75

Case: Starbucks

- 35,000+ stores, $32 billion+ revenue; **vertically integrated**, working with about **400,000 coffee producers**; **C.A.F.E.** (Coffee and Farmer Equity) Practices. S76
- 2008: **Peter Gibbons** simplified the chain into four macro-operations (plan, source, manufacture, deliver), centralized logistics, and a binary **scorecard** (safety, on-time supply and fulfilment, total cost, financial performance). S77
- Results: **98.6%** ethically sourced coffee; sales up **5%**. S78

LECTURE 34 S82–S99

Achieving Supply Chain Strategic Fit

- **Competitive strategy** defines which customer needs a firm meets and how it stands apart: Wal-Mart (low prices, availability), McMaster-Carr (MRO convenience and responsiveness), Blue Nile vs Zales (variety and low price online vs in-store experience). S84–S85
- SC strategies: efficient, responsive, lean, agile, collaborative. S87–S88

> STRATEGIC FIT S89
>
> Consistency between the customer priorities of the competitive strategy and the supply chain capabilities specified by the supply chain strategy; both strategies share the same goals.

Three steps S91: (1) understand the customer and supply chain uncertainty · (2) understand supply chain capabilities · (3) achieve strategic fit (coordinate functional strategies).

Step 1: customer and uncertainty

- Customer needs: quantity per lot, variety, service level, tolerated response time, price, rate of innovation. S92
- Implied demand uncertainty **increases** when: quantity range grows, lead time falls, variety grows, channels grow, innovation rate grows, required service level grows. S93

ATTRIBUTE	LOW IMPLIED UNCERTAINTY	HIGH IMPLIED UNCERTAINTY
Product margin	Low	High
Average forecast error	10%	40–100%
Average stockout rate	1–2%	10–40%
Forced season-end markdown	0%	10–25%

From S94.

- Supply uncertainty: higher for new products, lower for mature ones (Hau L. Lee, 2002). S95
- Spectrum: **salt at a supermarket** (predictable supply and demand) → **existing automobile model** → **new communication device** (highly uncertain). S96

Step 2: supply chain capabilities

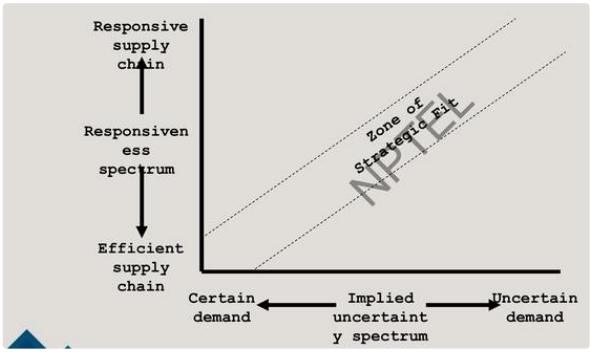
- **Responsiveness** = ability to respond to wide quantity ranges, meet short lead times, handle large variety, build innovative products, meet high service level, handle supply uncertainty. S97
- **Cost–responsiveness efficient frontier**: more responsiveness costs more. S98
- Responsiveness spectrum: **integrated steel mill** (highly efficient) → **Hanes apparel** (somewhat efficient) → **most automotive production** (somewhat responsive) → **Dell** (highly responsive). S99

LECTURE 35 S100–S133

Achieving Strategic Fit (contd.)

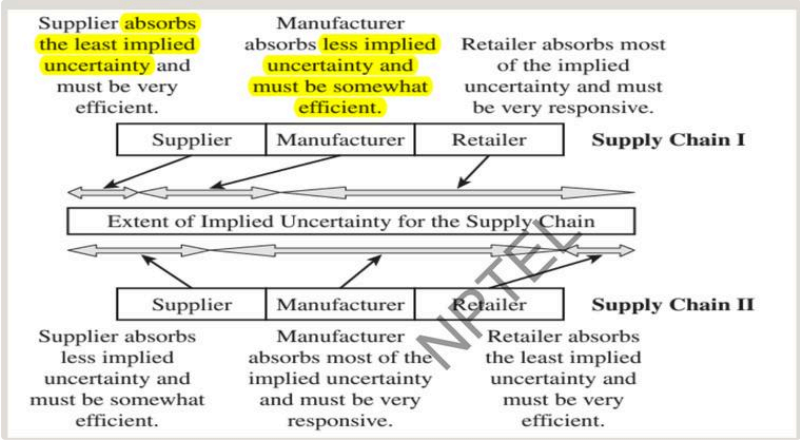
Step 3: achieve fit

- Match responsiveness to implied uncertainty: **high uncertainty → responsive**, **low → efficient**. McMaster-Carr chooses a responsive chain for MRO variety and fast delivery. S102
- Assign roles across stages: a retailer with little inventory passes uncertainty to a flexible manufacturer (England, Inc.). S103



Original slide S104. The **zone of strategic fit** runs diagonally: certain demand ↔ efficient chain, uncertain demand ↔ responsive chain.

- Two extremes: efficient (**Walmart**) and responsive (**Dell**). There is **no right SC strategy independent of competitive strategy**, but there is a right one for a given competitive strategy. S105



Original slide S106. Chain I: the retailer absorbs most uncertainty and is very responsive; the supplier is very efficient. Chain II: the manufacturer absorbs most and is very responsive; the retailer is very efficient.

	EFFICIENT	RESPONSIVE
Primary goal	Lowest cost	Quick response
Product design	Minimize product cost	Modularity to allow **postponement**
Pricing	Lower margins	Higher margins
Manufacturing	High utilization	Capacity flexibility
Inventory	Minimize inventory	Buffer inventory
Lead time	Reduce, but not at greater cost	Aggressively reduce even if costly
Supplier selection	Cost and low quality	Speed, flexibility, quality
Transportation	Low-cost modes	Fast modes

Marshall L. Fisher, HBR 1997 S107.

- Other issues: competitive changes over time, **product life cycle**, multiple products and segments. S108
- Life cycle: early = uncertain demand, high margins, availability matters; late = predictable, low margins, price matters. The chain moves **from responsive to efficient** (pharma, Intel). S110 S114
- **Tailored supply chain**: efficient where implied uncertainty is low, responsive where high. Dell (contract makers for Walmart channel), Levi Strauss (standard vs custom jeans), **W.W. Grainger** (fast movers decentralized, slow movers centralized). S112–S113

Scope of strategic fit

Intracompany intraoperation	**Minimize local cost** view: each stage optimizes alone. S116
Intracompany intrafunctional	**Minimize functional cost** view. S116
Intracompany interfunctional	**Maximize company profit** view. S117

Intercompany interfunctional	**Maximize supply chain surplus** view (e.g. joint promotions). S117
Agile (flexible) intercompany	Keep fit while partners change over time. S118

- Challenges: growing product variety and shrinking life cycles (Apple limits variety), globalization (Honda's flexible plants), fragmented ownership, changing technology (Dell's move into retail), environment and sustainability (Starbucks). S121–S123
- **Walmart**: strategic sourcing, long-term vendor alliances, **cross-docking** at the heart of replenishment; out-of-stocks down **16%**. S124–S126
- **Dell**: before 2007, direct sales, build-to-order, JIT, supplier integration; after 2007, retail distribution, diversification, lean, partnerships, sustainability. S127–S129

LECTURE 36 S134–S159

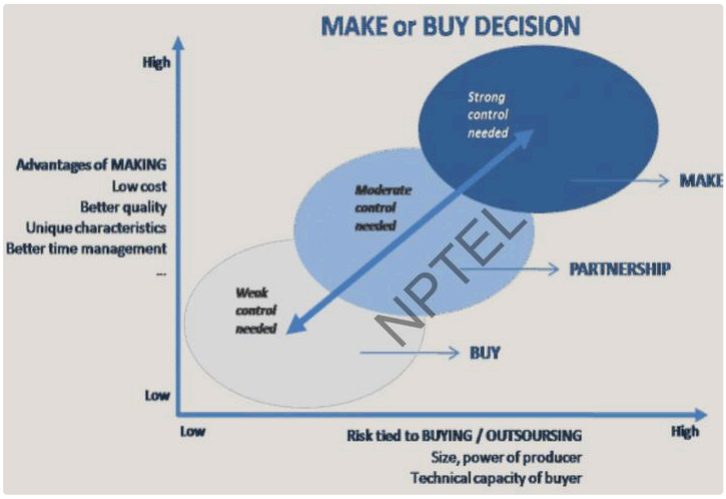
Make or Buy Decision

- A **strategic alliance** is a legal agreement between two or more companies committing resources to common goals (new markets, IP, infrastructure, technology, people). S136

TYPE	COMMITMENT · EXAMPLE S138–S140
Non-equity alliance	**Lowest**: no ownership shared. Two tech firms co-develop software.
Equity alliance	**Moderate**: one acquires equity in the other. A carmaker takes a stake in a battery firm.
Joint venture	**Highest**: a new co-owned legal entity (NewCo). Two pharma firms launch a drug.

- **Vertical** complementary alliances: firms at **different stages** of the value chain. **Horizontal**: firms at the **same stage**, often competitors. S141–S142
- Motives: domestic (market share, pooling resources, economies of scale, complementary resources); international (new markets, avoiding import barriers, international synergies, core competencies). S143
- Outsourcing benefits: cost savings, specialized expertise, **focus on core competencies**, scalability, efficiency, risk sharing, customer focus. S147–S148
- Limitations: loss of control, quality risk, communication challenges, dependency, data security, hidden costs, impact on internal workforce. S149–S150

COST OF MAKING (IN-HOUSE)	COST OF BUYING (OUTSOURCING) S151
Production, extra labour, monitoring, storage, waste disposal	Purchase price, sales tax, shipping, inventory holding, ordering



Original slide S152. Vertical axis: advantages of making (low cost, quality, uniqueness, time). Horizontal: risk of buying (producer power, buyer's technical capacity). Strong control needed → **make**; moderate → **partnership**; weak → **buy**.

- Implementing make-or-buy: assess the need → analyse cost and expertise → evaluate quality and control → consider capacity and scalability → case study → tips. S153
- Cases: **Apple** designs in the US, makes in China (manufacturing is not its core competency); **Nike** keeps design and marketing, uses contract manufacturers; **Google** core competencies in search algorithm, data analytics, UX design. S154–S156

BEFORE THE EXAM ALL LECTURES

Cheat sheet

3 + 2 impellers	Empowered customer, IT tools, globalization + sustainability, collaboration S4–S5
4 concepts	Systems, total cost, trade-off, bullwhip S7
5 business-level strategies	Cost leadership, differentiation, focused cost, focused differentiation, integrated S32
VRIN	Valuable, rare, costly to imitate, non-substitutable S58
Strategic fit steps	Customer/uncertainty → capabilities → match S91
Spectrum	Steel mill → Hanes → autos → Dell S99
5 scopes	Intraoperation → intrafunctional → interfunctional → intercompany → agile S115
Alliances	Non-equity < equity < joint venture S137

NUMBER	MEANING	SLIDE
$14.24 bn · 30→5	Unilever savings · warehouses to hubs	S47
10% · 10% · 5%	Unilever CPFR: forecast accuracy · inventory · sales	S48
1979	Porter's Five Forces published	S68
8.39	Amazon inventory turnover (2022)	S65
400,000 · 98.6%	Starbucks producers · ethically sourced coffee	S76–S78
10% vs 40–100%	Forecast error: low vs high implied uncertainty	S94
16%	Walmart out-of-stock reduction	S126

MEMORY HOOK (MY OWN, NOT FROM THE SLIDES)

"Uncertain needs speed, certain needs thrift": high implied uncertainty → responsive chain; low → efficient chain.

READ ONCE SOURCE CHECK

Slide quirks to watch

- The efficient-vs-responsive table lists efficient supplier selection as "cost and low quality" (Chopra's original wording). S107
- The scope slide names the fifth category "Flexible Interfunctional Scope" (S115) but describes it as "Agile Intercompany Scope" (S118).
- The Porter slide's numbering is shifted: "Buyer Power" appears as a bullet under item 1 and its details under item 2. S70
- "TAV" is listed as an IT tool with no expansion. S4
- Video or link-only slides: S6, S27, S86. Several title-only slides (S29, S51–S52, S80–S81, S133).

SELF-TEST 50 QUESTIONS

Week 7 practice questions

Answer these first, then check the key on the next pages. Option letters match the online quiz.

Q1 LECTURE 31

Which are the three traditional impellers of the supply chain?

A Empowered customer, developments in IT tools, globalization
B Plan, source, make
C Sustainability, collaboration, pricing
D Cost, quality, time

Q2 LECTURE 31

The phenomenon where small changes in consumer demand cause amplified order variations upstream is the:

A systems concept
B total cost concept
C bullwhip effect
D trade-off concept

Q3 LECTURE 31

Which is listed as a strategy to mitigate the bullwhip effect?

A Longer lead times
B Order batching
C Vendor-managed inventory
D Withholding information

Q4 LECTURE 31

Viewing the supply chain as one interconnected whole rather than isolated parts is the:

A systems concept
B trade-off concept
C total cost concept
D bullwhip concept

Q5 LECTURE 31

Which is NOT one of the primary value chain activities on the slide?

A Human resources
B Marketing and sales
C Distribution
D New product development

Q6 LECTURE 31

In the essential considerations for SC success, 'operations that make things happen, such as manufacturing and asset management' describe:

A Supply sense
B Demand sense
C Supply response
D Deciding and committing

Q7 LECTURE 31

The capability to sense and anticipate changes in customer demand accurately using data and analytics is:

A Supply sense
B Demand response
C Demand sense
D Supply response

Q8 LECTURE 31

Which IT tools are named under the IT impeller?

A GPS, RFID, UAV
B CRM, SRM, ISCM
C SCOR, DMAIC, PDCA
D MRP, ERP, VMI, TAV

Q9 LECTURE 32

Best-in-class companies, per the slides, have:

A one supply chain strategy for everything
B a unique supply chain strategy for each core function, e.g. make-to-stock, make-to-order, engineer-to-order
C only a pricing strategy
D no written strategy

Q10 LECTURE 32

Which is NOT a component of supply chain strategy on the slide?

A Logistics
B Advertising
C Sourcing
D Customer service

Q11 LECTURE 32

Serving a narrow market segment at the lowest cost is which business-level strategy?

A Differentiation
B Integrated strategy
C Focused cost leadership
D Cost leadership

Q12 LECTURE 32

Which characteristic is NOT listed for an effective supply chain strategy?

A Collaborative
B Isolated
C Agile
D Resilient

Q13 LECTURE 32

Unilever's CPFR stands for:

A Cost, Price, Freight, Revenue
B Collaborative Planning, Forecasting and Replenishment
C Central Purchasing For Retailers
D Continuous Process Flow Review

Q14 LECTURE 32

Through CPFR, Unilever improved forecast accuracy by:

A 10%
B 5%
C 14%
D 30%

Q15 LECTURE 32

Unilever merged almost 30 warehouses into how many distribution hubs?

A Fifteen
B Ten
C Five
D Two

Q16 LECTURE 32

Intel's ethical sourcing strategy uses a smelter audit protocol to ensure which metals are conflict-free?

A Copper, nickel, zinc, iron
B Silver, cobalt, lithium, platinum
C Tin, tungsten, gold, tantalum
D Aluminium, steel, titanium, lead

Q17 LECTURE 32

Competitive advantage requires a firm to decide:

A only location
B only what it will do
C what it will do and what it will not do
D only pricing

Q18 LECTURE 33

By studying the external environment, firms identify what they:

A must stop doing
B already do
C might choose to do
D can do

Q19 LECTURE 33

Which is NOT one of the four criteria of sustainable competitive advantage?

A Non-substitutable
B Valuable
C Rare
D Cheap to imitate

Q20 LECTURE 33

In SWOT, which two factors are external?

A Strengths and weaknesses
B Opportunities and threats
C Strengths and opportunities
D Weaknesses and threats

Q21 LECTURE 33

Porter's Five Forces model was first published in:

A 1979
B 1999
C 1989
D 1969

Q22 LECTURE 33

Which is NOT one of Porter's five forces?

- **A** Rivalry among competitors
- **B** Government subsidy
- **C** Supplier power
- **D** Threat of substitutes

Q23 LECTURE 33

Amazon's reported inventory turnover for FY ending December 2022 was about:

- **A** 8.39
- **B** 39.7
- **C** 16.6
- **D** 2.4

Q24 LECTURE 33

Starbucks' C.A.F.E. Practices stands for:

- **A** Central Africa Farmer Exchange
- **B** Coffee Audit For Ethics
- **C** Coffee and Farmer Equity
- **D** Cost And Freight Efficiency

Q25 LECTURE 33

Through C.A.F.E. and integration, Starbucks ethically sources what share of its coffee?

- **A** 90%
- **B** 98.6%
- **C** 100%
- **D** 75%

Q26 LECTURE 34

Strategic fit means consistency between:

- **A** marketing and HR
- **B** customer priorities of the competitive strategy and capabilities of the supply chain strategy
- **C** suppliers and retailers
- **D** price and cost

Q27 LECTURE 34

What is the first step in achieving strategic fit?

- **A** Understanding supply chain capabilities
- **B** Coordinating functional strategies
- **C** Choosing a 3PL
- **D** Understanding the customer and supply chain uncertainty

Q28 LECTURE 34

Which change in customer need INCREASES implied demand uncertainty?

- **A** Required service level increases
- **B** Lead time increases
- **C** Variety decreases
- **D** Fewer channels

Q29 LECTURE 34

Products with HIGH implied uncertainty typically have an average forecast error of:

- **A** 0%
- **B** 40–100%
- **C** 1–2%
- **D** 10%

Q30 LECTURE 34

Which product sits at the 'predictable supply and demand' end of the uncertainty spectrum?

- **A** Salt at a supermarket
- **B** A new communication device
- **C** A new drug
- **D** An existing automobile model

Q31 LECTURE 34

On the responsiveness spectrum, which is shown as highly responsive?

- **A** Most automotive production
- **B** Hanes apparel
- **C** Dell
- **D** Integrated steel mill

Q32 LECTURE 34

McMaster-Carr's competitive strategy centres on:

- **A** convenience, availability and responsiveness for MRO products
- **B** lowest price commodity goods
- **C** fast fashion
- **D** luxury diamonds

Q33 LECTURE 34

The cost-responsiveness efficient frontier shows that:

- **A** responsiveness is free
- **B** cost and responsiveness are unrelated
- **C** higher responsiveness comes with higher cost
- **D** lower cost raises responsiveness

Q34 LECTURE 35

For high implied uncertainty, the supply chain should target:

- **A** longest lead times
- **B** responsiveness
- **C** lowest inventory
- **D** efficiency

Q35 LECTURE 35

In an efficient supply chain, product design strategy aims to:

- **A** use buffer stock
- **B** use modularity to allow postponement
- **C** maximize variety
- **D** minimize product cost

Q36 LECTURE 35

A responsive supply chain's inventory strategy is to:

- **A** hold buffer inventory
- **B** minimize inventory
- **C** hold only raw materials
- **D** hold no inventory

Q37 LECTURE 35

As a product moves through its life cycle, the supply chain typically shifts from:

- **A** efficient to responsive
- **B** responsive to efficient
- **C** push to pull
- **D** lean to agile

Q38 LECTURE 35

W.W. Grainger stores fast-moving, low-uncertainty items:

- **A** in a centralized warehouse
- **B** at suppliers only
- **C** nowhere; made to order
- **D** in decentralized locations near customers

Q39 LECTURE 35

The 'maximize supply chain surplus' view corresponds to which scope of strategic fit?

- **A** Intracompany intrafunctional
- **B** Intracompany interfunctional
- **C** Intercompany interfunctional
- **D** Intracompany intraoperation

Q40 LECTURE 35

Through communication networks with suppliers, Walmart cut out-of-stock occurrences by:

- **A** 26%
- **B** 6%
- **C** 16%
- **D** 10%

Q41 LECTURE 35

Per the slides, which statement about SC strategy is correct?

- **A** There is one right supply chain strategy for every firm
- **B** Efficient chains are always better
- **C** There is no right SC strategy independent of competitive strategy
- **D** Responsive chains are always better

Q42 LECTURE 35

Before 2007, Dell's supply chain relied on:

- **A** direct sales and build-to-order
- **B** retail distribution
- **C** 4PL management
- **D** mass production for stock

Q43 LECTURE 36

Which strategic alliance involves the HIGHEST resource commitment?

- **A** Joint venture
- **B** Non-equity alliance
- **C** Equity alliance
- **D** Licensing

Q44 LECTURE 36

A car manufacturer acquiring a stake in a battery technology company is an example of:

- **A** non-equity alliance
- **B** countertrade
- **C** horizontal outsourcing
- **D** equity strategic alliance

Q45 LECTURE 36

Alliances between firms at different stages of the value chain are:

- **A** franchises
- **B** vertical complementary alliances
- **C** horizontal complementary alliances
- **D** joint ventures only

Q46 LECTURE 36

Which is a motive for INTERNATIONAL strategic alliances on the slide?

A Pooling resources
B Avoiding import barriers
C Economies of scale
D Market share

Q47 LECTURE 36

Which is a cost of BUYING (outsourcing) a product?

A Waste product disposal costs
B Ordering costs
C Production costs
D Monitoring costs

ANSWERS 50 QUESTIONS

Answer key

Q	ANS	WHY · SLIDE
Q1	A	Some literature adds sustainability and collaboration/integration (S5). S4
Q2	C	S11. S11
Q3	C	Better forecasting, shorter lead times, VMI, information sharing. S11
Q4	A	S8. S8
Q5	A	HR is a supporting function, with finance, accounting and IT. S13–S15
Q6	C	S17. S17
Q7	C	Demand response (S19) is reacting to those changes. S18
Q8	D	S4. S4
Q9	B	S25. S25
Q10	B	Components: sourcing, logistics, production, distribution, customer service. S28
Q11	C	Broad + lowest cost = cost leadership; narrow + lowest cost = focused cost leadership. S32
Q12	B	Collaborative, agile, resilient, digital, invest in cybersecurity. S37–S38
Q13	B	S47. S47
Q14	A	10% forecast accuracy, 10% less inventory holding, 5% more sales volume. S48
Q15	C	S47: five huge hubs. S47
Q16	C	S49, with RFID, bagging and tagging. S49
Q17	C	S31. S31
Q18	C	Internal environment → what they can do (S57). S56
Q19	D	The criterion is costly to imitate. S58
Q20	B	S67 diagram: internal (top) vs external (bottom). S67
Q21	A	Michael Porter, Harvard, 1979. S68
Q22	B	Supplier power, buyer power, new entrants, substitutes, rivalry. S69–S72
Q23	A	S65. S65
Q24	C	S76. S76
Q25	B	S78; sales rose 5%. S78
Q26	B	S89. S89
Q27	D	Step 2: capabilities; Step 3: achieve fit. S91
Q28	A	Higher service level, shorter lead time, more variety, more channels, faster innovation all raise implied uncertainty. S93
Q29	B	Low uncertainty: 10% forecast error, 1–2% stockout; high: 40–100%, 10–40%. S94
Q30	A	S96. S96
Q31	C	Steel mill: highly efficient; Hanes: somewhat efficient; autos: somewhat responsive; Dell: highly responsive. S99
Q32	A	S84. S84
Q33	C	S98. S98
Q34	B	High uncertainty → responsive; low → efficient. S102
Q35	D	Responsive chains use modularity to allow postponement. S107
Q36	A	S107. S107
Q37	B	Early: uncertain demand, high margins; late: predictable demand, price matters. S110
Q38	D	Slow movers go to a centralized warehouse. S113
Q39	C	Intraoperation = minimize local cost; intrafunctional = minimize functional cost; interfunctional = maximize company profit. S117
Q40	C	Cross-docking is at the heart of Walmart's replenishment. S126
Q41	C	But there is a right SC strategy for a given competitive strategy. S105
Q42	A	After 2007 it moved into retail channels. S127
Q43	A	A JV creates a new co-owned legal entity (NewCo). S140
Q44	D	S139. S139
Q45	B	Horizontal alliances are between firms at the same stage. S141
Q46	B	International: new markets, avoiding import barriers, synergies, core competencies. S143
Q47	B	Buying: purchase price, sales tax, shipping, inventory holding, ordering. S151
Q48	A	Moderate control → partnership; weak control → buy. S152
Q49	A	S154. S154

Q	ANS	WHY · SLIDE
Q50	**B**	Others: quality risk, communication, dependency, data security, hidden costs, workforce impact. S149

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